135. The average salary of 100 employees in an office is Rs. 16,000 per month. The management decided to raise salary of every employee by 5% but stopped a transport allowance of Rs. 800 per month which was paid earlier to every employee. What will be the new average monthly salary?

(a) Rs. 16,000
(b) Rs. 16,500
(c) Rs. 16,800
(d) Cannot be known since data are insufficient

Ans. (a) : Since, salary of each employee is increased by 5%, so net average increase in salary = 5%.
∴ New average monthly salary
= 16000 (1+ 5/100)
= 16800
But stopped transport allowance 800 per month
Then average salary of 100 employ
= 16800 – 800 =16000
Hence option (a) is correct.

136. A person has to completely put each of three liquids: 403 litres of petrol, 465 litres of diesel and 496 litres of Mobil Oil in bottles of equal size without mixing any of the above three types of liquids such that each bottle is completely filled. What is the least possible number of bottles required?

(a) 34 (b) 44
(c) 46 (d) None of the above

Ans. (b) : Maximum capacity of each bottle can be found by taking the H.C.F of the three given liquids.
Maximum capacity of each bottle = HCF of 403, 465 and 496 = 31
Number of bottles for 403 l of petrol = 403 / 31 = 13
Number of bottles for 465 l of diesel =465 / 31 = 15
Number of bottles for 496 l of mobile oil = 496 / 31=16
Hence, total number of bottles = 13 + 15 + 16 = 44

137. Amit starts from a point A and walks to another point B, and then returns from B to A by his car and thus takes a total time of 6 hours and 45 minutes. If he had driven both ways in his car, he would have taken 2 hours less. How long would it take for him to walk both ways?

(a) 7 hours 45 minutes
(b) 8 hours 15 minutes
(c) 8 hours 30 minutes
(d) 8 hours 45 minutes

Ans. (d) : Let the distance be x km.
(Time taken to walk x km) + (Time taken to drive x km) =27/4 hrs.
⇒ (Time taken to walk 2x km) + (Time taken to drive 2x km) =27/2 hrs.
But time taken to drive 2x km =19/4 hrs.
Therefore, time taken to walk 2x km is:
27/2−19/4=54/4−19/4=54−19/4 =35/4 =8 3/4
Hence, it would take 8 hours 45 minutes for him to walk both ways.

138. A and B can complete work together in 5 days. If A works at twice his speed and B at half of his speed, this work can b finished in 4 days. How many days would it take for A alone to complete the job?

(a) 10 (b) 12
(c) 15 (d) 18

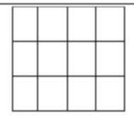
Ans. (a) : Let alone time for A and B be x and y respectively.
Now, $1/x + 1/y = 1/5$ .....(i)
Also, $2/x + 1/2y = 1/4$ .....(ii)
Solving (i) and (ii), we get, $x = 10$ and $y = 10$

139. If all the numbers from 501 to 700 are written, what is the total number of times does the digits 6 appear?

(a) 138 (b) 139
(c) 140 (d) 141

Ans. (c) : Numbers from 501 to 599 which have 6 as digit are 506, 516, 526, 536, 546, 556, 560, 561, 562, 563, 564, 565, 566, 567,568, 569, 576, 586 and 596, i.e., 6 occurs 20 times.
Number of times 6 occurs from 600 to 699 = 100 + 20 = 120
∴ Total number of times 6 occurs = 20 + 120 = 140

140.



In how many maximum different ways can 3 identical balls be placed in the 12 squares (each) ball to be placed in the exact centre of the squares and only one ball is to be placed in one square shown in the figure given above such that they do not lie along the same straight line?

(a) 144 (b) 200
(c) 204 (d) 216

Ans. (c) : 3 balls can be placed in any of the 12 squares in $^{12}C_3$ ways.
Total no. of arrangements = $^{12}C_3 = 220$
Now, assume that balls lie along the same line
There can be 3 cases

(i) When balls lie along the straight horizontal line. 3 balls can be put in any of the 4 boxes along the horizontal row in $^4C_3$ ways. Since there are 3 rows, so number of ways = $^4C_3 \times 3 = 12$
(ii) When balls lie along the vertical straight line 3 balls can be put in any of the 3 boxes along the vertical row in $^3C_3$ ways. Now as there are 4 vertical rows, so number of ways in 2nd case = $^3C_3 \times 4 = 4$
(iii) Balls lie along the 2 diagonal lines towards the left and 2 diagonal lines towards the right.
No. of ways = 2 + 2 = 4
∴ No. of ways when balls lie along the line = 12 + 4 + 4 = 20
∴ Number of ways when balls don't lie along the line = Total No. of ways – number of ways when ball lie along the line = 220 – 20 = 200.

141. Groups each containing 3 boys are to be formed out of 5 boys A, B, C, D and E such that no one group contains both C and D together. What is the maximum number of such different groups?

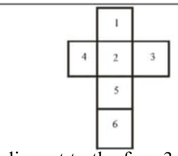
(a) 5 (b) 6
(c) 7 (d) 8

Ans. (c) : $5C_3$
Total number of arrangements, when any 3 boys are selected out of 5= $5C_3$
Now, when groups contain both C and D, then their selection is fixed and the remaining 1 boy can be selected out of the remaining 3 boys. It can be done in $3C_1$ ways.
So, number of groups, when none contains both C and D= total number of arrangements-number of arrangements when group contains both C and D = $^5C_3 - {}^3C_1 = 10–3 = 7$

142. Six faces of a cube are numbered from 1 to 6, each face carrying one different number. Further:
1. The face 2 is opposite to the face 6.
2. The face 1 is opposite to the face 5.
3. The face 3 is between the face 1 and the face 5
4. The face 4 is adjacent to the face 2.
Which one of the following is correct?

(a) The face 2 is adjacent to the face 3
(b) The face 6 is between the face 2 and the face 4
(c) The face 1 is between the face 5 and the face 6
(d) None of the above

Ans. (a) : It is very clear from figure.



The face 2 is adjacent to the face 3.

143. Each of the 3 persons is to be given some identical items such that product of the numbers of items received by each of the three persons is equal to 30. In how many maximum different ways can this distribution be done?

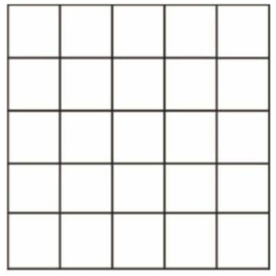
(a) 21 (b) 24
(c) 27 (d) 33

Ans. (c) : 30 = 2 × 3 × 5 and all the positive divisor of 30 are – 1, 2, 3, 5, 6, 10,15, 30
we have to find the number of triples (a, b, c) such that a × b × c = 30.
Now, 30 = 15 × 2 = 10 × 3 = 6 × 5 = 2 × 3 × 5. Each of these can be made into triples of divisors by introducing factors of 1 suitably if required. Thus we have the following triples–
30 = (30, 1, 1), (1, 30, 1) and (1, 1, 30) → 3
15 × 2 = 6 permutations of (1, 2, 15)
10 × 3 = 6 permutations of (1, 3, 10)
6 × 5 = 6 permutations of (1, 5, 6)
2 × 3 × 5 = 6 permutations of (2, 3, 5)
∴ altogether the distribution can be done in (4 × 6) + 3 = 27 ways.

144. 6 equidistant vertical lines are drawn on a board. 6 equidistant horizontal lines are also drawn on the board cutting the 6 vertical lines, and the distance between any two consecutive horizontal lines is equal to the between any two consecutive vertical lines. What is the maximum number of squares thus formed?

(a) 37 (b) 55
(c) 126 (d) 225

Ans. (b) :



Counting number of square from figure
Case 1 : Single square box = 5 × 5 = 25 .....(i)
Case 2 : Double square box = 4 × 4 = 16 .....(ii)
Case 3 : Triple square box = 3 × 3 = 9 .....(iii)
Case 4 : Square with 4 square = 2 × 2 = 4 .....(iv)
Case 5 : Square with 5 square box = 1 .....(v)
Adding all the square boxes from (i), (ii), (iii), (iv) and (v) We get
25 + 16 + 9 + 4 + 1 = 55 square boxes.

145.